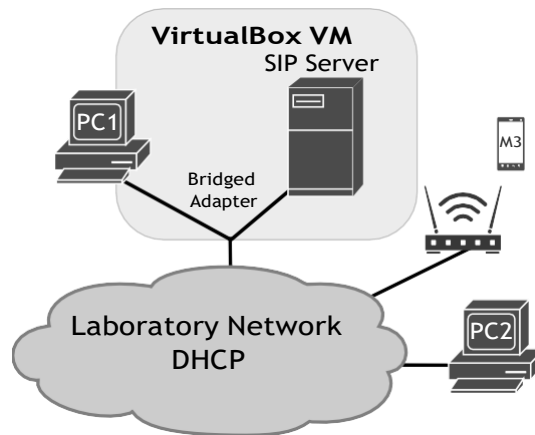


REDES DE COMUNICAÇÕES 2



SIP, RTP and RTCP Protocols

1. The SIP server must be implemented as a virtual Ubuntu/Debian server. Connect the VM to your PC (PC1) Wi-Fi/Ethernet interface using a Bridged Adapter.

Generate a different MAC address for your VM network interface:

In VirtualBox: Settings→ Network→ Adapter/Advanced→ Click the icon right of the MAC address field.

In QEMU/virt-manager: Not required.

Boot the SIP Server and identify the DHCP acquired IPv4 address (assuming from now on 192.168.220.x).

Identify the DHCP acquired IPv4 address of your PC1. From your PC1, test the connectivity with your SIP Server (using the command ping).

Note: You may connect to your server using SSH (Linux: `ssh labcom@192.168.220.x` , Windows: install a SSH client and connect to the server), in order to make future configurations easier by command line copy and paste.

Note2: The server should have installed the VoIP gateway and conference server (Asterisk) with default configuration.

Change to root terminal with the command `sudo su`.

Editing the `/etc/asterisk/sip.conf` file add a new user (to the end of the file):

```
[PintoDaCosta]
type=friend
host=dynamic
secret=labcom
context=phones
allow=all
```

And editing the `/etc/asterisk/extensions.conf` file add a new section (to the end of the file) with a phone extension (2000) to deliver a welcome message:

```
[phones]
exten => 2000,1,Answer(500)
exten => 2000,n,Playback(demo-congrats)
exten => 2000,n,Playback(vm-goodbye)
exten => 2000,n,Hangup()
```

Restart Asterisk: `service asterisk restart`

2. In your PC1, start a VoIP softphone ([linphone](#)) and login to your SIP server account with the credentials defined before

In, HOME → ACCOUNT ASSISTANT → USE A SIP ACCOUNT

Username: PintoDaCosta
SIP Domain: 192.168.220.x
Transport: UDP

or in, Preferences → SIP accounts → ADD ACCOUNT

SIP address: sip:PintoDaCosta@192.168.220.x
SIP Server address: <sip:192.168.220.x;transport=udp>
Transport: UDP

On top left status/name region, choose the active account to the one in your server. Start a Wireshark capture **on the server VM**.

On your VoIP client Preferences → SIP accounts → Edit → Register (toggle), unregistered and register in SIP server a couple of times.

>> Analyze the exchanged SIP messages and identify the purpose of the REGISTER messages (and if present, SUBSCRIBE and PUBLISH messages).

3. Start new Wireshark capture **on the server VM** and change the status in your VoIP client (top left) to BUSY. Change the status back to AVAILABLE. Analyze the exchanged SIP messages.

>> Identify the purpose of the PUBLISH messages (based on the XML contents of the message body).

4. Start new Wireshark capture, make an audio call to extension 2000 (sip:2000@192.168.220.x) for a welcome message from server, and wait for the end of the message:

>> Analyze the exchanged SIP request and status messages and identify the purpose of the following request messages: INVITE, ACK and BYE.

Note: If the call has no sound some codecs may be missing. To solve the problem install VLC in your client (Android by default has the codecs).

5. Editing the /etc/asterisk/sip.conf file add a new user:

```
[RuiCosta]
type=friend
host=dynamic
secret=labcom
context=phones
allow=all
```

Editing the /etc/asterisk/extensions.conf file define extension numbers for all users:

```
[phones]
...
exten => 2001,1,Dial(SIP/PintoDaCosta,10)

exten => 2002,1,Dial(SIP/RuiCosta,10)
```

Restart Asterisk: service asterisk restart

6. Using a second PC (PC2), or a mobile phone (M3) with the linphone app installed, or making arrangements with a neighbor group, register the second user with your Asterisk server. On your PC1, start new Wireshark capture and make an audio call to the second user extension (2002@192.168.220.x). After a few seconds hang up the call. Analyze the exchanged SIP request and status messages.

>> Identify the purpose of the following SIP messages: INVITE, ACK, BYE, Trying, Ringing and OK.

EXERCÍCIO 2

Ip do PC : 192.168.1.202 = src
IP da VM : 192-168.1.198 = dst = server
Request - URI = server
Max Forwards = 70 = n° max de vezes que o pacote poderá ser reencaminhado
Pacote Register do SIP :

Wireshark - Packet 24 - enp0s3

▶ Frame 24: 847 bytes on wire (6716 bits), 847 bytes captured (6716 bits) on interface 0

▶ Ethernet II, Src: a4:42:3b:80:cd:ac (a4:42:3b:80:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)

▶ Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198

▶ User Datagram Protocol, Src Port: 5060, Dst Port: 5060

Session Initiation Protocol (REGISTER)

▶ Request-Line: REGISTER sip:192.168.1.198 SIP/2.0

Method: REGISTER

▶ Request-URI: sip:192.168.1.198

[Resent Packet: False]

Message Header

▶ Via: SIP/2.0/UDP 192.168.1.202:5060;branch=z9hG4bK.ZHACFIAM4;rport

▶ From: <sip:PintoDaCosta@192.168.1.198>;tag=6y0n38Rbv

▶ To: sip:PintoDaCosta@192.168.1.198

▶ CSeq: 22 REGISTER

Call-ID: 4e268bhzlu

Max-Forwards: 70

Supported: replaces, outbound, gruu

Accept: application/sdp

Accept: text/plain

Accept: application/vnd.gsm.a-rcs-ft-http+xml

▶ Contact: <sip:PintoDaCosta@192.168.1.202;transport=udp>;sip.instance="urn:uuid:0efa8025-691b-000f-8269-816129056cd7"

Expires: 3600

User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 2009, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cdd0918e

▶ Authorization: Digest realm="asterisk", nonce="268612c1", algorithm=MD5, username="PintoDaCosta", uri="sip:192.168.1.198", response="f554e0f73de4d33b1857d8a554450b34"

Pacote Publish do SIP :

▶ Frame 2968: 1061 bytes on wire (8488 bits), 1061 bytes captured (8488 bits) on interface 0

▶ Ethernet II, Src: a4:42:3b:80:cd:ac (a4:42:3b:80:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)

▶ Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198

▶ User Datagram Protocol, Src Port: 5060, Dst Port: 5060

Session Initiation Protocol (PUBLISH)

▶ Request-Line: PUBLISH sip:PintoDaCosta@192.168.1.198 SIP/2.0

Method: PUBLISH

▶ Request-URI: sip:PintoDaCosta@192.168.1.198

[Resent Packet: False]

Message Header

▶ Via: SIP/2.0/UDP 192.168.1.202:5060;branch=z9hG4bK.07M1ta2or;rport

▶ From: <sip:PintoDaCosta@192.168.1.198>;tag=6Au8r0cEh

▶ SIP from address: sip:PintoDaCosta@192.168.1.198

▶ SIP from tag: 6Au8r0cEh

▶ To: sip:PintoDaCosta@192.168.1.198

▶ SIP to address: sip:PintoDaCosta@192.168.1.198

▶ CSeq: 20 PUBLISH

Call-ID: v9Axc6D00K

Max-Forwards: 70

Supported: replaces, outbound, gruu

Event: presence

Content-Type: application/pidf+xml

Content-Length: 511

Expires: 3600

User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 2009, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cdd0918e

Message Body

EXERCÍCIO 3

▶ Frame 11: 1116 bytes on wire (8928 bits), 1116 bytes captured (8928 bits) on interface 0

▶ Ethernet II, Src: a4:42:3b:80:cd:ac (a4:42:3b:80:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)

▶ Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198

▶ User Datagram Protocol, Src Port: 5060, Dst Port: 5060

Session Initiation Protocol (PUBLISH)

▶ Request-Line: PUBLISH sip:PintoDaCosta@192.168.1.198 SIP/2.0

Method: PUBLISH

▶ Request-URI: sip:PintoDaCosta@192.168.1.198

[Resent Packet: False]

Message Header

▶ Via: SIP/2.0/UDP 192.168.1.202:5060;branch=z9hG4bK.hfrqtqc06;rport

▶ From: <sip:PintoDaCosta@192.168.1.198>;tag=gKgeMuJPa

▶ To: sip:PintoDaCosta@192.168.1.198

▶ CSeq: 20 PUBLISH

Call-ID: Hn7r69b4w4

Max-Forwards: 70

Supported: replaces, outbound, gruu

Event: presence

Content-Type: application/pidf+xml

Content-Length: 566

Expires: 3600

User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 2009, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cdd0918e

Content Body

eXtensible Markup Language

▶ Frame 278: 1061 bytes on wire (8488 bits), 1061 bytes captured (8488 bits) on interface 0

▶ Ethernet II, Src: a4:42:3b:80:cd:ac (a4:42:3b:80:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)

▶ Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198

▶ User Datagram Protocol, Src Port: 5060, Dst Port: 5060

Session Initiation Protocol (PUBLISH)

▶ Request-Line: PUBLISH sip:PintoDaCosta@192.168.1.198 SIP/2.0

Method: PUBLISH

▶ Request-URI: sip:PintoDaCosta@192.168.1.198

[Resent Packet: False]

Message Header

▶ Via: SIP/2.0/UDP 192.168.1.202:5060;branch=z9hG4bK.DGA3ztRK-r;rport

▶ From: <sip:PintoDaCosta@192.168.1.198>;tag=3YVw6zdwj

▶ To: sip:PintoDaCosta@192.168.1.198

▶ CSeq: 20 PUBLISH

Call-ID: g059g0h-3-

Max-Forwards: 70

Supported: replaces, outbound, gruu

Event: presence

Content-Type: application/pidf+xml

Content-Length: 511

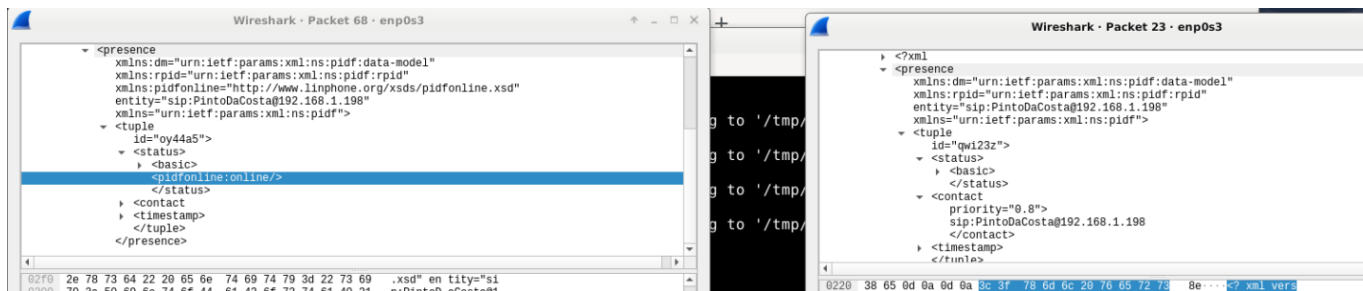
Expires: 3600

User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 2009, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cdd0918e

Message Body

eXtensible Markup Language

Pacote Publish quando estás no modo "busy" não aparece o parametro sublinhado, aparece apenas quando estás no modo "online":



EXERCÍCIO 4

18	17	313769287	192.168.1.202	192.168.1.198	SIP/SDP	1254 Request: INVITE sip:20000@192.168.1.198
19	17	314318836	192.168.1.198	192.168.1.202	SIP	568 Status: 401 Unauthorized
20	17	351988637	192.168.1.202	192.168.1.198	SIP	494 Request: ACK sip:20000@192.168.1.198
21	17	352275115	192.168.1.202	192.168.1.198	SIP/SDP	1429 Request: INVITE sip:20000@192.168.1.198
22	17	355950610	192.168.1.202	192.168.1.202	SIP	503 Status: 100 Trying
23	17	359872952	192.168.1.198	192.168.1.202	SIP/SDP	997 Status: 200 OK
39	17	573823468	192.168.1.202	192.168.1.198	SIP	597 Request: ACK sip:20000@192.168.1.198:5060
3246	49	635661341	192.168.1.198	192.168.1.202	SIP	633 Request: BYE sip:PintoDaCosta@192.168.1.202;transport=udp
3249	49	888522497	192.168.1.202	192.168.1.198	SIP	415 Status: 200 OK

Pacote Invite:

Frame 18: 1254 bytes on wire (10032 bits), 1254 bytes captured (10032 bits) on interface 0
Ethernet II, Src: aa:42:3b:80:cd:ac (aa:42:3b:80:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)
Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198
User Datagram Protocol, Src Port: 5060, Dst Port: 5060
Session Initiation Protocol (INVITE)
Request-Line: INVITE sip:20000@192.168.1.198 SIP/2.0
Method: INVITE
Request-URI: sip:20000@192.168.1.198
[Resent Packet: False]
Message Header
Via: SIP/2.0/UDP 192.168.1.202:5060;branch=29h64bk.T-vosX-2;rport
From: <sip:PintoDaCosta@192.168.1.198>;tag=13a5r0gep
SIP From address: sip:PintoDaCosta@192.168.1.198
SIP From tag: 13a5r0gep
To: sip:20000@192.168.1.198
SIP to address: sip:20000@192.168.1.198
CSeq: 20 INVITE
Sequence Number: 20
Method: INVITE
Call-ID: IaLM02PKld
Max-Forwards: 70
Supported: replaces, outbound, gruu
Allow: INVITE, ACK, CANCEL, OPTIONS, BYE, REFER, NOTIFY, MESSAGE, SUBSCRIBE, INFO, PRACK, UPDATE
Content-Type: application/sdp
Content-Length: 527
Contact: <sip:PintoDaCosta@192.168.1.202;transport=udp;expires=3600>;sip.instance="urn:uuid:0efa8025-691b-000f-8269-816129b56cd7"
Contact-URI: sip:PintoDaCosta@192.168.1.202;transport=udp
Contact-URI User Part: PintoDaCosta
Contact-URI Host Part: 192.168.1.202
Contact-URI parameter: transport=udp
Contact parameter: expires=3600
Contact parameter: sip.instance="urn:uuid:0efa8025-691b-000f-8269-816129b56cd7"
User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 20H9, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cdd0918e
Message Body
Session Description Protocol
Session Description Protocol Version (v): 0
Owner/Creator, Session Id (o): PintoDaCosta 165 1928 IN IP4 192.168.1.202
Owner Username: PintoDaCosta
Session ID: 165
Session Version: 1928
Owner Network Type: IN
Owner Address Type: IP4
Owner Address: 192.168.1.202
Session Name (s): Talk
Connection Information (c): IN IP4 192.168.1.202
Time Description, active time (t): 0 0
Session Attribute (a): rtcp-xr:rcvr-rtt=all:10000 stat-summary=loss,dup,jitt,TTL voip-metrics
Media Description, name and address (m): audio 7078 RTP/AVP 96 97 98 0 8 18 99 100 101
Media Attribute (a): rtmap:96 opus/48000/2
Media Attribute (a): fmtp:96 useinbandfec=1
Media Attribute (a): rtmap:97 speex/16000
Media Attribute (a): fmtp:97 vbr=on
Media Attribute (a): rtmap:98 speex/8000
Media Attribute (a): fmtp:98 vbr=on
Media Attribute (a): fmtp:18 annex=bvyes
Media Attribute (a): rtmap:99 telephone-event/48000
Media Attribute (a): rtmap:100 telephone-event/16000
Media Attribute (a): rtmap:101 telephone-event/8000
Media Attribute (a): rtcp-fb:* trr-int 1000
Media Attribute (a): rtcp-fb:* ccm tsmbr

Pacote Trying:

Frame 44/5: 585 bytes on wire (4680 bits), 585 bytes captured (4680 bits) on interface 0
Ethernet II, Src: PcsCompu_b8:da:cd (08:00:27:b8:da:cd), Dst: aa:42:3b:80:cd:ac (aa:42:3b:80:cd:ac)
Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198
User Datagram Protocol, Src Port: 5060, Dst Port: 5060
Session Initiation Protocol (100)
Status-Line: SIP/2.0 100 Trying
Status-Code: 100
[Resent Packet: False]
[Request Frame: 44/41]
[Response Time (ms): 25]
Message Header
Via: SIP/2.0/UDP 192.168.1.202:5060;branch=29h64bk.6PCBT5-3Q;received=192.168.1.202;rport=5060
From: <sip:PintoDaCosta@192.168.1.198>;tag=duFILTXY-
SIP From address: sip:PintoDaCosta@192.168.1.198
SIP From tag: duFILTXY-
To: sip:20000@192.168.1.198
SIP to address: sip:20000@192.168.1.198
Call-ID: HfES5v22k
CSeq: 21 INVITE
Server: Asterisk PBX 16.2.1-0fsg-1deb10u2
Allow: INVITE, ACK, CANCEL, OPTIONS, BYE, REFER, SUBSCRIBE, NOTIFY, INFO, PUBLISH, MESSAGE
Supported: replaces, timer
Contact: <sip:20000@192.168.1.198:5060>
Contact-URI: sip:20000@192.168.1.198:5060
Contact-URI User Part: 20000
Contact-URI Host Part: 192.168.1.198
Contact-URI Port: 5060
Content-Length: 0

Pacote ACK:

Frame 448: 587 bytes on wire (4696 bits), 587 bytes captured (4696 bits) on interface 0
Ethernet II, Src: aa:42:3b:80:cd:ac (08:00:27:b8:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)
Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198
User Datagram Protocol, Src Port: 5060, Dst Port: 5060
Session Initiation Protocol (ACK)
Request-Line: ACK sip:20000@192.168.1.198:5060 SIP/2.0
Method: ACK
Request-URI: sip:20000@192.168.1.198:5060
[Resent Packet: False]
[Request Frame: 41/41]
[Response Time (ms): 253]
Message Header
Via: SIP/2.0/UDP 192.168.1.202:5060;rport;branch=29h64bk.E3y73Ahv-
From: <sip:PintoDaCosta@192.168.1.198>;tag=duFILTXY-
SIP From address: sip:PintoDaCosta@192.168.1.198
SIP From tag: duFILTXY-
To: sip:20000@192.168.1.198;tag=asf158f2
SIP to address: sip:20000@192.168.1.198
SIP to tag: asf158f2
CSeq: 21 ACK
Sequence Number: 21
Method: ACK
Call-ID: HfES5v22k
Max-Forwards: 70
Authorization: Digest realm="asterisk", nonce="67b136c4", algorithm=MD5, username="PintoDaCosta", uri="sip:20000@192.168.1.198", response="a4c1473f78f7a5c30c7a7570e63508b"
Authentication Scheme: Digest
Realm: "asterisk"
Nonce Value: "67b136c4"
Algorithm: MD5
Username: "PintoDaCosta"
Authentication URI: "sip:20000@192.168.1.198"
Digest Authentication Response: "a4c1473f78f7a5c30c7a7570e63508b"
User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 20H9, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cdd0918e

Pacote OK:

```

> Frame 4476: 979 bytes on wire (7832 bits), 979 bytes captured (7832 bits) on interface 0
> Ethernet II, Src: PcsCompu_b8:da:cd (08:00:27:b8:da:cd), Dst: a4:42:3b:80:cd:ac (a4:42:3b:80:cd:ac)
> Internet Protocol Version 4, Src: 192.168.1.198, Dst: 192.168.1.202
> User Datagram Protocol, Src Port: 5060, Dst Port: 5060
> Session Initiation Protocol (200)
  - Status-Line: SIP/2.0 200 OK
    Status-Code: 200
    [Resent Packet: False]
    [Request Frame: 4474]
    [Response Time (ms): 36]
  - Message Header
    - Via: SIP/2.0/UDP 192.168.1.202:5060;branch=z9hG4bK.6pCBT5-3Q;received=192.168.1.202;rport=5060
    - From: <sip:PintoDaCosta@192.168.1.198>;tag=duFILTXN-
    - SIP from address: sip:PintoDaCosta@192.168.1.198
    - SIP from tag: duFILTXN-
    - To: sip:2000@192.168.1.198;tag=as5f158ef2
    - SIP to address: sip:2000@192.168.1.198
    - SIP to tag: as5f158ef2
    - Call-ID: -HRE5sv2Zk
    - CSeq: 21 INVITE
    - Sequence Number: 21
    - Method: INVITE
    - Server: Asterisk PBX 16.2.1-dfsg-1-deb10u2
    - Allow: INVITE, ACK, CANCEL, OPTIONS, BYE, REFER, SUBSCRIBE, NOTIFY, INFO, PUBLISH, MESSAGE
    - Supported: replaces, timer
    - Contact: <sip:2000@192.168.1.198:5060>
    - Contact URI: sip:2000@192.168.1.198:5060
      Contact URI User Part: 2000
      Contact URI Host Part: 192.168.1.198
      Contact URI Host Port: 5060
    - Content-Type: application/sdp
    - Content-Length: 432
  - Message Body
    - Session Description Protocol
      - Session Description Protocol Version (v): 0
      - Owner/Creator, Session Id (o): root 1286988738 1286988738 IN IP4 192.168.1.198
        Owner Username: root
        Session ID: 1286988738
        Session Version: 1286988738
        Owner Network Type: IN
        Owner Address Type: IP4
        Owner Address: 192.168.1.198
      - Session Name (s): Asterisk PBX 16.2.1-dfsg-1-deb10u2
      - Connection Information (c): IN IP4 192.168.1.198
      - Time Description, active time (t): 0 0
      - Media Description, name and address (m): audio 12052 RTP/AVP 0 8 18 98 97 96 101
      - Media Attribute (a): rtmap:8 PCMA/8000
      - Media Attribute (a): rtmap:18 G729/8000
      - Media Attribute (a): fftp:18 annexbno
      - Media Attribute (a): rtmap:98 speex/8000
      - Media Attribute (a): rtmap:97 speex/16000
      - Media Attribute (a): rtmap:99 opus/48000/2
      - Media Attribute (a): rtmap:101 telephone-event/8000
      - Media Attribute (a): fftp:101 0-16
      - Media Attribute (a): maxptime:60
      - Media Attribute (a): sendrecv
      - Media Description, name and address (m): video 0 RTP/AVP 96

```

Pacote BYE

```

> Frame 4953: 602 bytes on wire (4816 bits), 602 bytes captured (4816 bits) on interface 0
> Ethernet II, Src: a4:42:3b:80:cd:ac (a4:42:3b:80:cd:ac), Dst: PcsCompu_b8:da:cd (08:00:27:b8:da:cd)
> Internet Protocol Version 4, Src: 192.168.1.202, Dst: 192.168.1.198
> User Datagram Protocol, Src Port: 5060, Dst Port: 5060
> Session Initiation Protocol (BYE)
  - Request-Line: BYE sip:2000@192.168.1.198:5060 SIP/2.0
    Method: BYE
    Request-URI: sip:2000@192.168.1.198:5060
    [Resent Packet: False]
  - Message Header
    - Via: SIP/2.0/UDP 192.168.1.202:5060;branch=z9hG4bK.3QWqEIVv;rport
    - From: <sip:PintoDaCosta@192.168.1.198>;tag=duFILTXN-
    - SIP from address: sip:PintoDaCosta@192.168.1.198
    - SIP from tag: duFILTXN-
    - To: <sip:2000@192.168.1.198>;tag=as5f158ef2
    - SIP to address: sip:2000@192.168.1.198
    - SIP to tag: as5f158ef2
    - CSeq: 22 BYE
    - Sequence Number: 22
    - Method: BYE
    - Call-ID: -HRE5sv2Zk
    - User-Agent: Linphone Desktop/4.4.1 (diogopires) Windows 10 Version 2009, Qt 5.15.2 LinphoneCore/5.1.19-1-g6cd09919e
  - Authentication Scheme: Digest
    Realm: "asterisk"
    Nonce Value: "67d136c4"
    Algorithm: MD5
    Username: "PintoDaCosta"
    Authentication URI: "sip:2000@192.168.1.198:5060"
    Digest Authentication Response: "80bd75eed79f67aa8dafb141312677e"

```

EXERCÍCIO 6



```

> Session Initiation Protocol (180)
  - Status-Line: SIP/2.0 180 Ringing
    Status-Code: 180
    [Resent Packet: False]
    [Request Frame: 33]
    [Response Time (ms): 401]
  - Message Header
    - Via: SIP/2.0/UDP 192.168.1.198:5060;branch=z9hG4bK1133e33e
      Transport: UDP
      Sent-by Address: 192.168.1.198
      Sent-by port: 5060
      Branch: z9hG4bK1133e33e
    - From: <sip:PintoDaCosta@192.168.1.198>;tag=as2bac9c1e
      SIP from address: sip:PintoDaCosta@192.168.1.198
      SIP from tag: as2bac9c1e
    - To: <sip:RuiCosta@192.168.1.207:46043;transport=udp>;tag=y1-kiAk
      SIP to address: sip:RuiCosta@192.168.1.207:46043;transport=udp
      SIP to tag: y1-kiAk
    - Call-ID: 3802efc151baedce3fce2d9a0f8f6b6e@192.168.1.198:5060
    - CSeq: 102 INVITE
    - Sequence Number: 102
    - Method: INVITE
    - User-Agent: LinphoneAndroid/4.6.10 (T774H) LinphoneSDK/5.1.42 (tags/5.1.42^0)
    - Supported: replaces, outbound, gruu

```